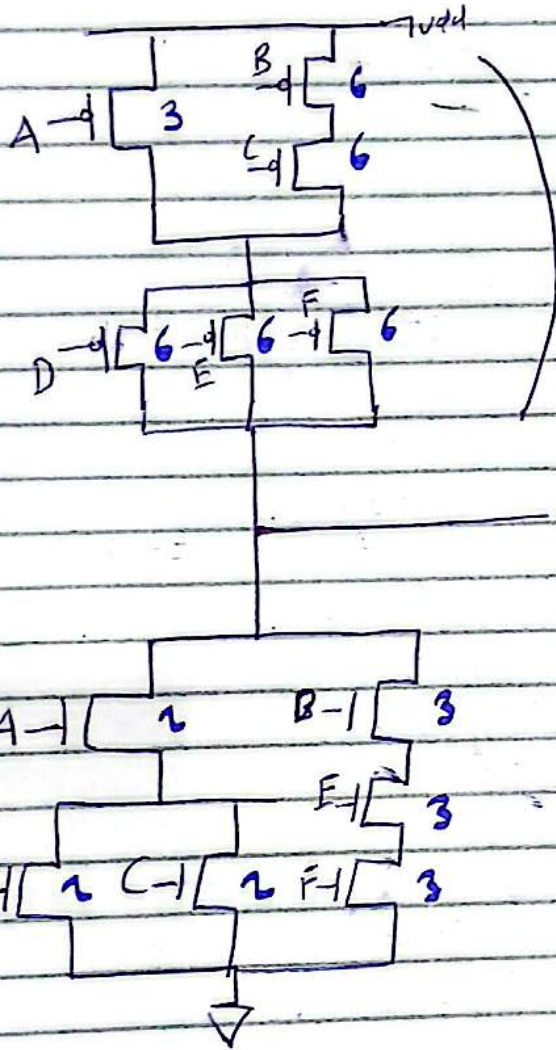


VLSI

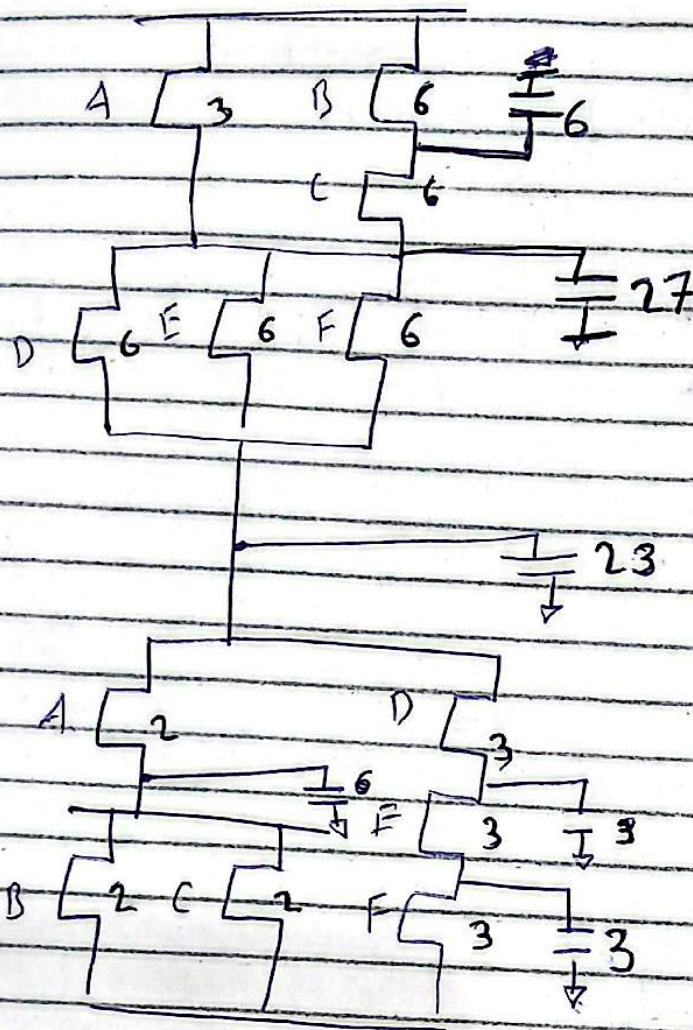
Assignment 2

$$F = (A \cdot (B + C) + D \cdot E \cdot F)$$



$$R \quad \frac{6B}{K} = 1K \Rightarrow K=6$$

$$\frac{4B}{K} = 13 = K=4$$

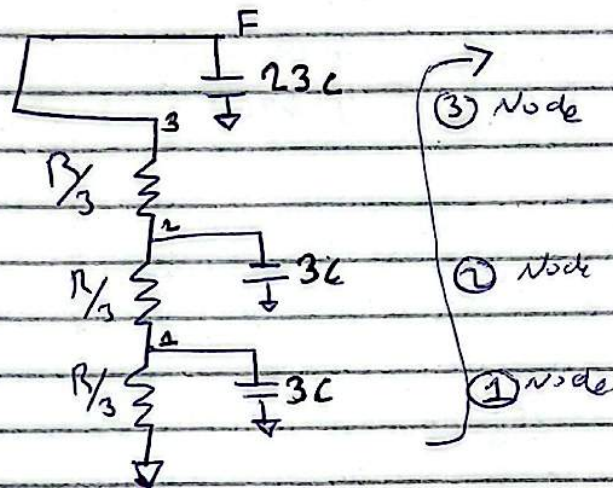


$\Rightarrow$  Worst case is Maximum Delay  
 $\Rightarrow$  out of all parallel paths only '1' is on



Falling

$$A = 0 \quad B = 0 \quad C = 0 \quad (D \quad E \quad F) = 1$$



Elmore Delay path.

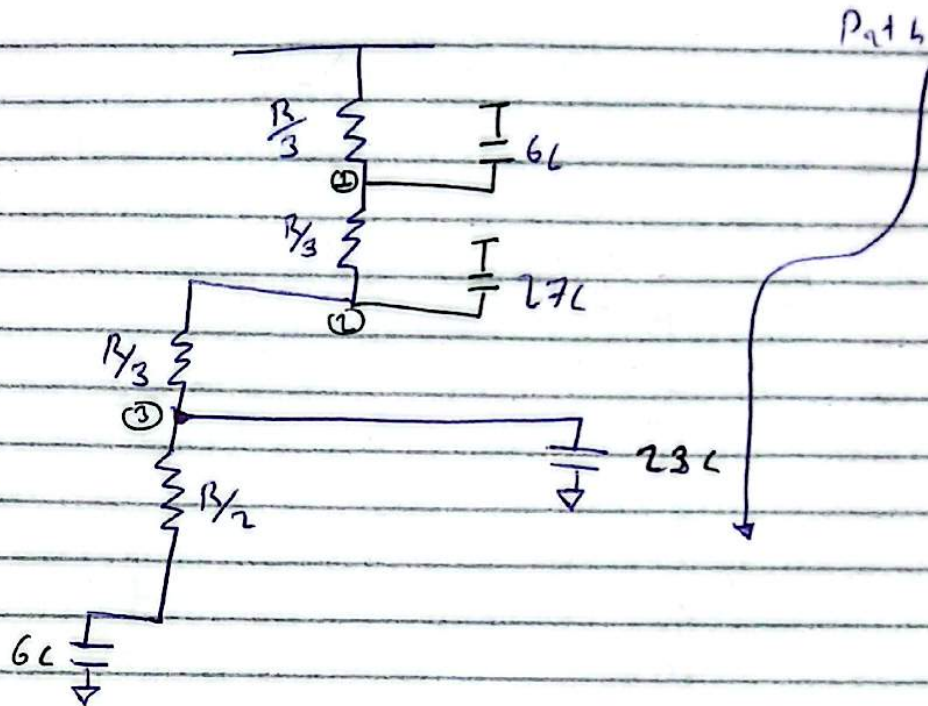
$\tau_{\text{falling}} =$

$$\frac{R}{3} (3C) + \left( \frac{R}{3} + \frac{R}{3} \right) 3C + \left( \frac{R}{3} + \frac{R}{3} + \frac{R}{3} \right) 3C$$

=

Rising

$$A = 1 \quad B = 0 \quad C = 0 \quad D = 0 \quad E = 1 \quad F = 1$$



Source is  $V_{dd}$  in Rising

Resistance is short in  $= \frac{R}{2}$

$$\tau_{\text{rising}} = \frac{R}{3}(6C) + \frac{2R}{3}(27C) + \frac{3R}{3}(23C) + \frac{3R}{3}(6C)$$